

Solutions

Question 1

(a)

$$\frac{d}{dt} \left(\sqrt{t^2 - 3t} \right) = \boxed{\frac{1}{2} (t^2 - 3t)^{-1/2} (2t - 3)}$$

(b) Write as a power:

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{\sqrt{t^2 - 3t}} \right) &= \frac{d}{dt} (t^2 - 3t)^{-1/2} \\ &= \boxed{-\frac{1}{2} (t^2 - 3t)^{-3/2} (2t - 3)} \end{aligned}$$

(c)

$$\begin{aligned} \frac{d}{dx} \left((1 + e^{-x})^{-1} \right) &= - (1 + e^{-x})^{-2} (-e^{-x}) \\ &= \boxed{e^{-x} (1 + e^{-x})^{-2}} \quad \text{or} \quad \boxed{\frac{e^{-x}}{(1 + e^{-x})^2}} \end{aligned}$$

(d)

$$\frac{d}{dt} (\ln(8 - t^3)) = \boxed{\frac{1}{8 - t^3} (-3t^2)}$$

(e)

$$\begin{aligned} &\ln(\cos^2(x) + 1) \cdot (e^x + \sin(x))^3 \\ &\frac{d}{dx} \left[\ln(\cos^2(x) + 1) \cdot (e^x + \sin(x))^3 \right] \\ &= \boxed{\frac{1}{\cos^2 x + 1} (-2 \cos x \sin x) (e^x + \sin(x))^3 + \ln(\cos^2(x) + 1) \left[3(e^x + \sin(x))^2 (e^x + \cos(x)) \right]} \end{aligned}$$

No further simplification is required.

Question 2

(a) Rewrite before differentiating.

$$\begin{aligned} r(t) &= \frac{5t^4 - 2t}{\sqrt{t}} \\ &= 5t^{7/2} - 2t^{1/2} \end{aligned}$$

$$\boxed{r'(t) = \frac{35}{2} t^{5/2} - t^{-1/2}}$$

(b)

$$r(t) = \frac{\sin(7t)}{t^3} \quad \text{This needs the quotient rule, or rewrite with } t^{-3}:$$

$$r'(t) = \frac{7 \cos(7t) (t^3) - (\sin(7t)) (3t^2)}{t^6}$$

Optional tidying of t powers:

$$= \frac{7 \cos(7t)}{t^3} - \frac{3 \sin(7t)}{t^4}$$

Question 3

$$\vec{a} = \langle 1, 2, 3 \rangle, \quad \vec{b} = \langle 0, -1, 1 \rangle, \quad \vec{c} = \langle 2, 4, 6 \rangle$$

(a) $\vec{a} + \vec{b} = \langle 1, 1, 4 \rangle$

(b)

$$\begin{aligned} \vec{b} - 3\vec{c} &= \langle 0, -1, 1 \rangle - \langle 6, 12, 18 \rangle \\ &= \langle -6, -13, -17 \rangle \end{aligned}$$

(c)

$$\|\vec{a} - \vec{b}\| = \|\langle 1, 3, 2 \rangle\| = \sqrt{1^2 + 3^2 + 2^2} = \sqrt{14}$$

(d)

$$\vec{b} \cdot \vec{c} = 0 - 4 + 6 = 2$$

Question 4

(a) $\vec{a} = \langle 1, k, 3 \rangle, \quad \vec{b} = \langle 2, -3, 4 \rangle.$

The vectors are perpendicular if $\vec{a} \cdot \vec{b} = 0$. Set

$$\begin{aligned} \vec{a} \cdot \vec{b} &= 2 - 3k + 12 = 0 \\ 14 &= 3k \end{aligned}$$

$$\boxed{k = \frac{14}{3}}$$

(b) $\vec{c} = \langle q, 2 \rangle, \quad \vec{d} = \langle 2, 3 \rangle.$

They are parallel if $\vec{c} = k \vec{d}$ or $\langle q, 2 \rangle = k \langle 2, 3 \rangle.$

By component,

$$\begin{aligned}q &= 2k \\ 2 &= 3k \Rightarrow k = \frac{2}{3}\end{aligned}$$

so

$$\boxed{q = 2 \left(\frac{2}{3}\right) = \frac{4}{3}}$$

Alternatively, by inspection: going from $\vec{c}$ to $\vec{d}$ multiplies the second component by $\frac{3}{2}$, so going back divides by $\frac{3}{2}$ (i.e. multiplies by $\frac{2}{3}$). Hence $q = 2 \cdot \frac{2}{3}$, giving $\boxed{q = \frac{4}{3}}$.

Question 5

Given $\langle -3, 4 \rangle$. We want a perpendicular vector. Let $\langle u, v \rangle$ be our vector.

$$\begin{aligned}\langle -3, 4 \rangle \cdot \langle u, v \rangle &= 0 \Rightarrow -3u + 4v = 0 \\ &\Rightarrow 3u = 4v \\ &\Rightarrow \boxed{u = \frac{4}{3}v}\end{aligned}$$

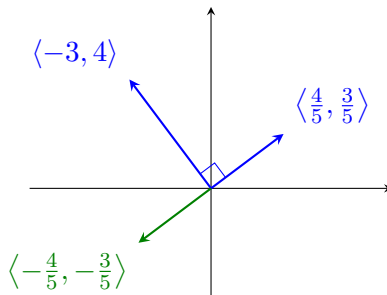
Pick e.g. $v = 1 \Rightarrow \langle \frac{4}{3}, 1 \rangle$ is perpendicular to $\langle -3, 4 \rangle$.

It's just not a unit vector, but that is easy to fix:

$$\hat{a} = \frac{1}{\sqrt{\left(\frac{4}{3}\right)^2 + 1^2}} \langle \frac{4}{3}, 1 \rangle = \frac{3}{5} \langle \frac{4}{3}, 1 \rangle = \boxed{\langle \frac{4}{5}, \frac{3}{5} \rangle}$$

Is this the only answer? No!

Sketch:



$\boxed{\langle -\frac{4}{5}, -\frac{3}{5} \rangle}$ is another unit vector perpendicular to $\langle -3, 4 \rangle$.

Follow-up: For any vector $\langle a, b \rangle$ in 2D, the vector

$$\langle b, -a \rangle$$

will be perpendicular.

- exchange the coordinates, and
- flip one of the signs.

Question 6

We want

$$\langle 10, -4, 73 \rangle \cdot \langle a, b, c \rangle = 0.$$

Simplest answer is $\langle a, b, c \rangle = \langle 0, 0, 0 \rangle$.